

Introduction to Modeling for Formal Verification

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Today's goals



▼

Module 1

Introduction to Model Checking

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Module 2

Modeling

▼

Module 3

Modeling System Composition

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- Modeling Concurrency Using Transition Systems
- ✓

Concurrency and Interleaving

Video • 10 min
- ✓

Concurrency and Interleaving: Examples

Video • 8 min
- ✓

Principles of Model Checking - Section 2.2.1, (Page 35-39)

Reading • 30 min
- ✓

Modeling Concurrency Using Transition Systems Quiz

Graded Assignment • Grade: 100%

- Modeling Communications Via Shared Variables
- ✓

Interleaving Program Graphs

Video • 11 min
- ✓

Critical Actions and Atomicity

Video • 10 min
- ✓

Peterson's Mutual Exclusion Algorithm

Video • 12 min
- ✓

Principles of Model Checking - Section 2.2.2, (Page 39-47)

Reading • 30 min
- ✓

Modeling Communications Via Shared Variables Quiz

Graded Assignment • Grade: 100%

- Modeling Communications Via Handshaking
- ✓

Synchronization and Handshaking

Video • 12 min
- ✓

Example: Railroad Crossing

Video • 13 min
- ✓

Principles of Model Checking - Section 2.2.3, (Page 48-52)

Reading • 30 min
- ✓

Modeling Communications Via Handshaking Quiz

Graded Assignment • Grade: 100%

- Synchronous Parallelism
- ✓

Synchronous Product - Part 1

Video • 10 min
- ✓

Synchronous Product - Part 2

Video • 10 min
- ✓

The State Space Explosion Problem

Video • 9 min
- ✓

Principles of Model Checking - Section 2.2.6, (Page 73-75)

Reading • 20 min
- Synchronous Parallelism Quiz

Graded Assignment • 20 min

Principles of Model Checking - Section 2.2.6, (Page 73-75)

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Textbook - Principles of Model Checking - Chapter 2.2.6 (page 73-75)

Section 2.2.6 introduces synchronous parallelism, which is used for describing hardware sequential circuits that are comprised of components synchronized by a global clock signal. It also explains how to find the synchronous product of a set of transition systems of components that are synchronized by a clock signal. Finally, Section 2.3 discusses the state space explosion problem and explains its main sources. It also details the complexities of individual program graphs and the parallel composition of components that contribute to this problem.

Go to next item

✓ Completed

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